



East West University

CSE 475

Machine Learning

Task – 02

Group – 10

Submitted To:

Dr. Raihan Ul Islam
Associate professor,
Department of Computer Science and
Engineering East West University,
Dhaka

Submitted By:

Md. Rayhan Ahamed (2021-3-60-197)
Sadia Islam Dia (2022-3-60-062)
Nurtaz Ahmed (2020-2-60-062)
Tahmina Ahmed (2022-2-60-151)

Date of Submission: 19-08-2026

Executive Summary

This report details the system architecture, data representation, and implementation strategy for **Corr-GCN**—a graph-based deep learning framework designed for surface electromyography (sEMG) hand gesture classification. By converting high-density multi-channel sEMG feature vectors into dynamic graph structures based on inter-channel feature correlations, Corr-GCN models spatial dependencies across muscle groups to classify 8 discrete hand gestures.

Dataset & Graph Formulation

Data Preprocessing

The dataset comprises 619,479 windowed sEMG feature samples across 44 subjects (37 cross-validation subjects and 7 test subjects). Each sample contains 168 feature dimensions extracted across $N = 24$ spatial electrode channels and $M = 7$ feature metrics:

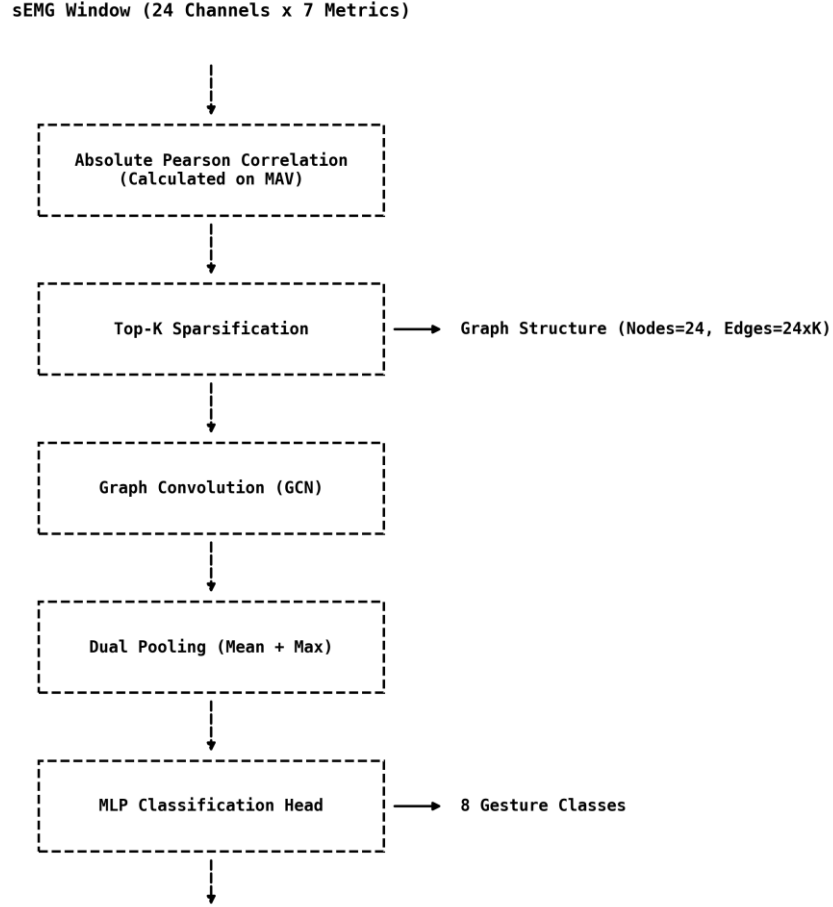
$$\{\text{Metrics}\} = \{\text{MAV}\}, \{\text{RMS}\}, \{\text{STD}\}, \{\text{VAR}\}, \{\text{WL}\}, \{\text{ZC}\}, \{\text{SSC}\}$$

Graph Structure

- **Nodes (V):** Represent the 24 sEMG channels ($|V| = 24$), where each node i holds a 7-dimensional feature vector $x_i \in \mathbb{R}^7$
- **Adjacency Matrix (A):** Built per training fold using the absolute Pearson correlation matrix $C \in \mathbb{R}^{24 \times 24}$ of the Mean Absolute Value (MAV) feature across channels:

$$C_{ij} = |\text{corr}(X_{i,\text{MAV}}, X_{j,\text{MAV}})|, C_{ii} = 0$$

- **Sparsification (Top-K):** To eliminate weak correlation noise, each node i retains undirected weighted edges only to its K most correlated neighbors, with edge weights $w_{ij} = C_{ij}$.



Network Architecture Specification

The baseline **CorrGCN** model processes PyTorch Geometric graph data objects containing node features $\mathbf{X} \in \mathbb{R}^{24 \times 7}$, edge indices, and normalized edge weights:

1. Graph Convolution Backbone:

Node features pass through sequential GCNConv layers with ReLU activations:

$$h_i^{(l+1)} = \text{ReLU} \left(\sum_j (w_{ij} / \sqrt{(\hat{D}_{ii} \hat{D}_{jj})}) W^{(l)} h_j^{(l)} \right)$$

Default channel dimensions transition from **7** $\rightarrow$ **64** $\rightarrow$ **128**.

2. Dual Readout Pooling:

Node-level representations are aggregated into a single graph embedding by concatenating global mean and global max pooling vectors:

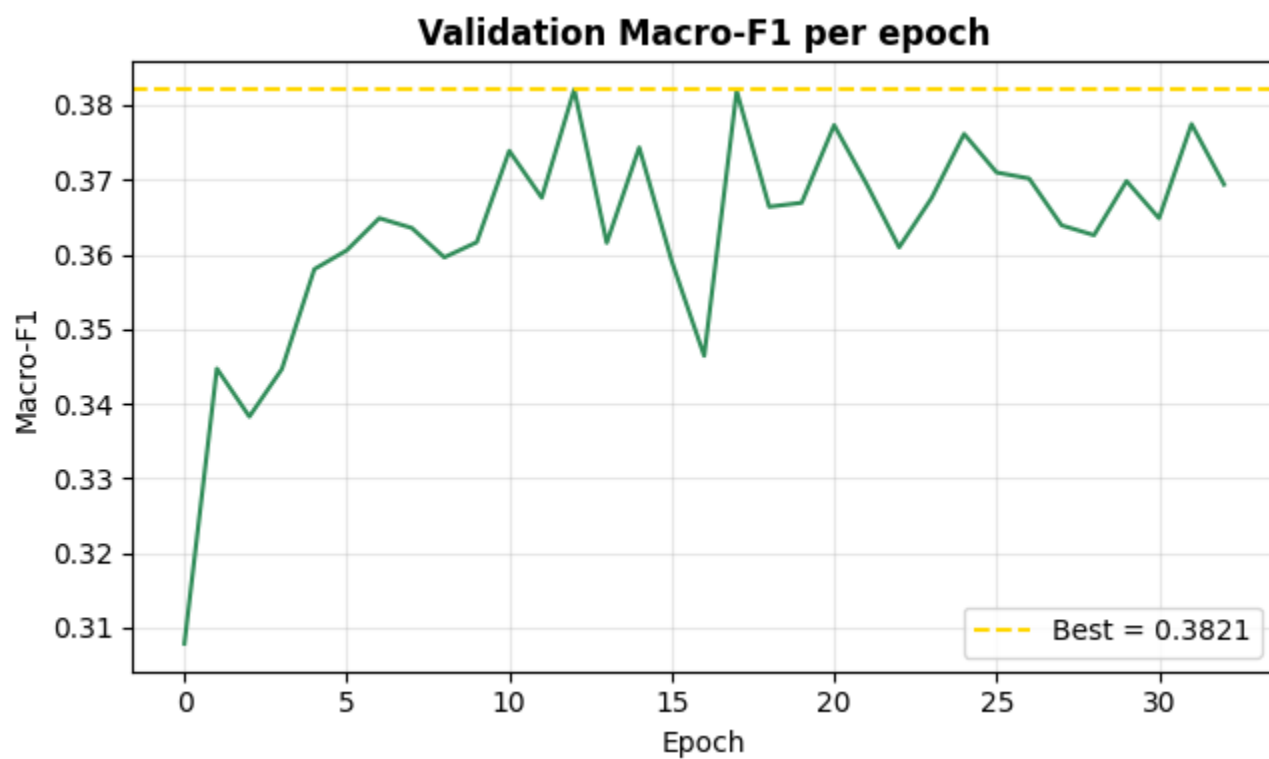
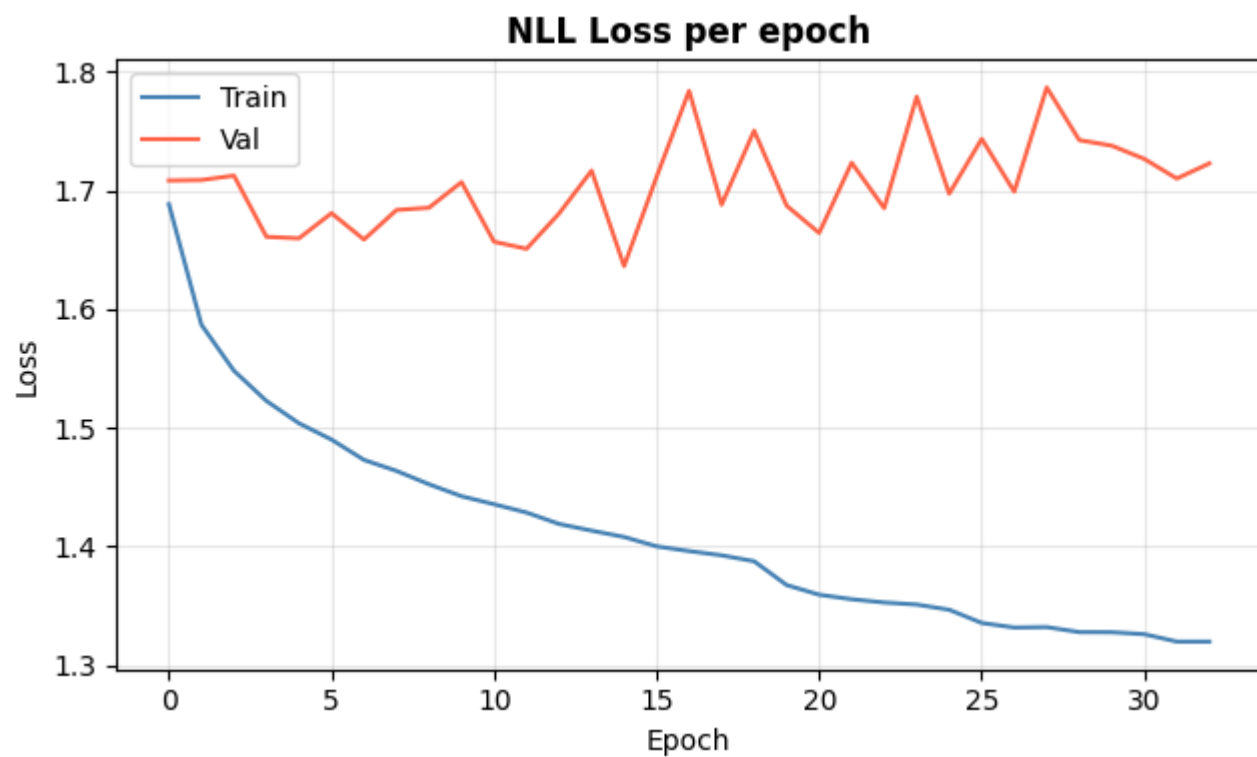
$$\mathbf{z}_{\text{graph}} = [\text{global_mean_pool}(\mathbf{H}), \text{global_max_pool}(\mathbf{H})] \in \mathbb{R}^{256}$$

3. Classification Head:

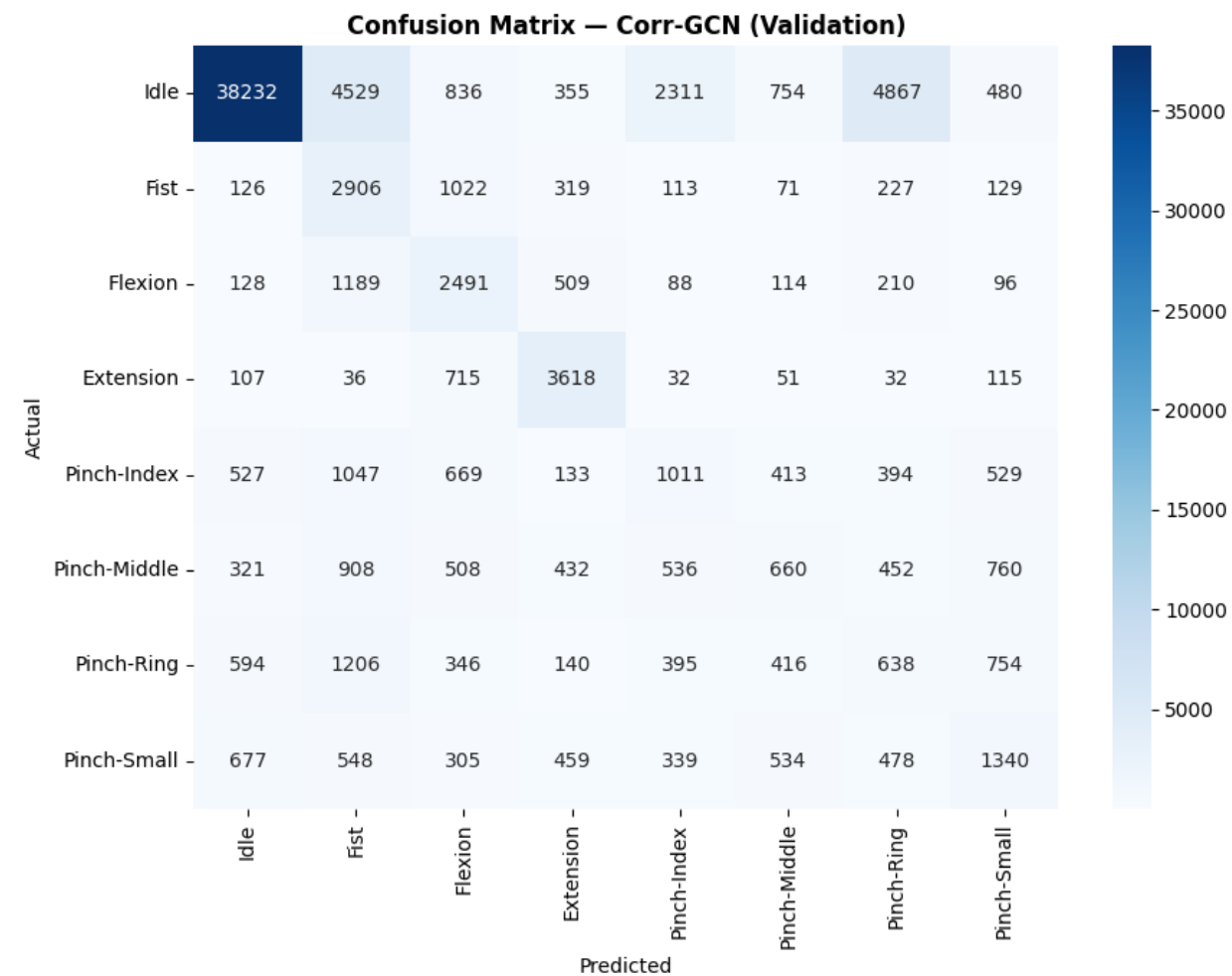
$\mathbf{z}_{\text{graph}} \rightarrow \text{Linear}(256, 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(256, 128) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.3) \rightarrow \text{Linear}(128, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 8)$

Training Protocol & Baseline Results

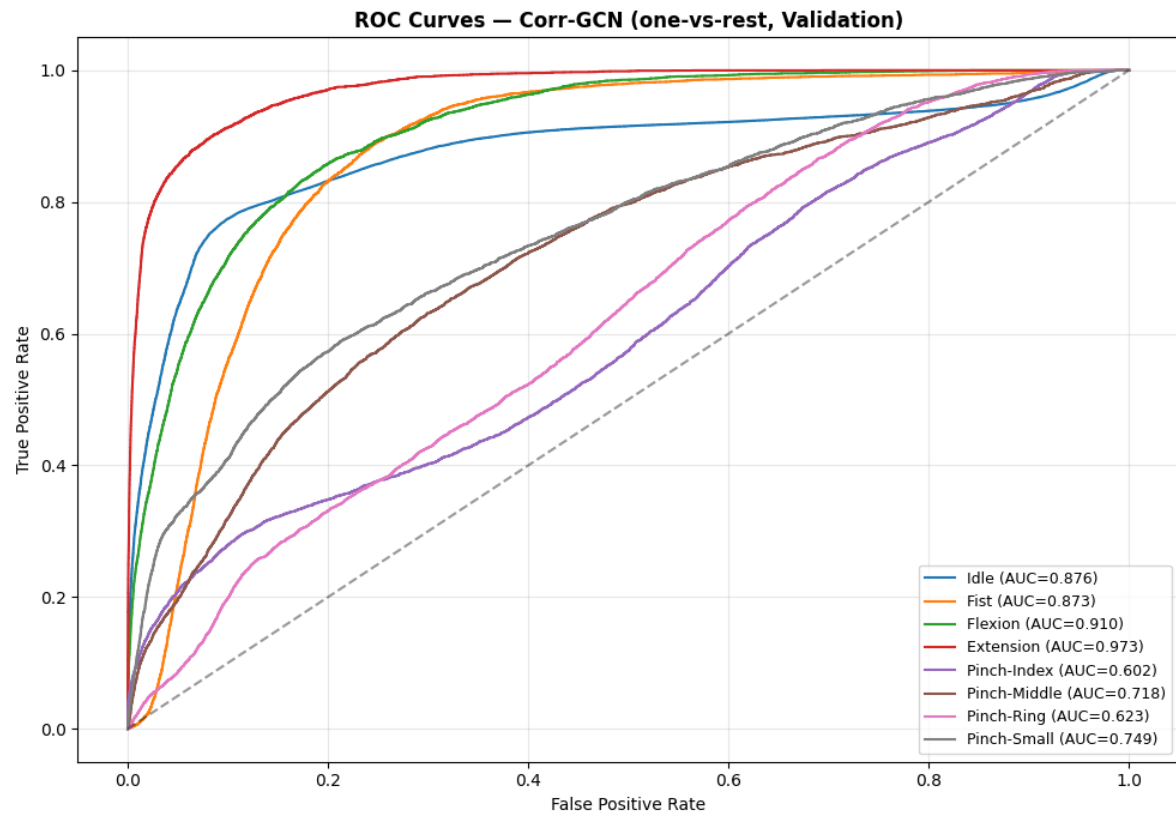
- **Validation Strategy:** 5-Fold `GroupKFold` split on 37 training subjects to evaluate cross-subject generalization.
- **Loss & Optimization:** Inverse class-weighted Negative Log-Likelihood (NLLLoss), optimized via AdamW ($\text{lr} = 10^{-3}$, $\text{weight_decay} = 10^{-4}$) with ReduceLROnPlateau scheduling.
- **Baseline Metrics:** The baseline 2-layer CorrGCN with $\text{Top-K} = 4$ and mean_max pooling achieved a 5-fold CV Macro-F1 score of **0.3993 $\pm$ 0.0199**.



Confusion Matrix



ROC Curves



Precision Recall Curves

